

Performance Testing

Date	4 July 2026
Team ID	SWTID-2026-71 47
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1 : Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Loan Prediction Module (/result API)
Test Environment	Local (Flask Web Application)
Test Date	4 July

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Low-risk applicant loan prediction	10	60	Prediction returned successfully within 2 seconds
2	High-risk applicant loan prediction	20	60	High-risk prediction displayed correctly
3	Multiple loan requests simultaneously	50	120	System handles concurrent requests without failure
4	Continuous prediction requests	100	120	Stable performance with low response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status(Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.4seconds	Pass	Within limit
3	Throughput (Req/sec)	> 40 Req/sec	52 Req/sec	Pass	Good throughput
4	Error Rate	< 1 %	0 %	Pass	No errors No errors
5	CPU Utilization	< 80%	58 %	Pass	Normal usage
6	Memory Utilization	< 80%	63 %	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application processed loan prediction requests quickly and accurately. The Flask application remained stable under multiple concurrent requests, with no significant performance degradation. This demonstrates that the application is reliable, efficient, and capable of handling multiple users simultaneously while maintaining consistent performance.

Bottlenecks Identified:

No major bottlenecks were observed during testing. A slight increase in response time occurred under higher load, but it remained within acceptable limits.

Optimization Steps Taken:

- Optimized Flask request handling.
- Reduced unnecessary processing during prediction.
- Improved model loading efficiency.
- Minimized response generation time

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

SMART LENDER

HomeAboutPredictionContact

AI Powered Loan Prediction

Check your loan eligibility quickly and securely.

Predict Now

About Smart Lender

Smart Lender is a web-based loan prediction system that helps users determine loan eligibility quickly based on applicant details.

Loan Prediction Form

Applicant Income

3000

Loan Amount

400

Credit History

Good

Predict Loan

Back to Home

Loan Prediction Result

✓

Loan Approved

Congratulations! Based on the details provided, your loan is eligible for approval.

Try AgainHome

